

Course: ENSF 480 – Principles of Software Design

Lab #: Lab 3

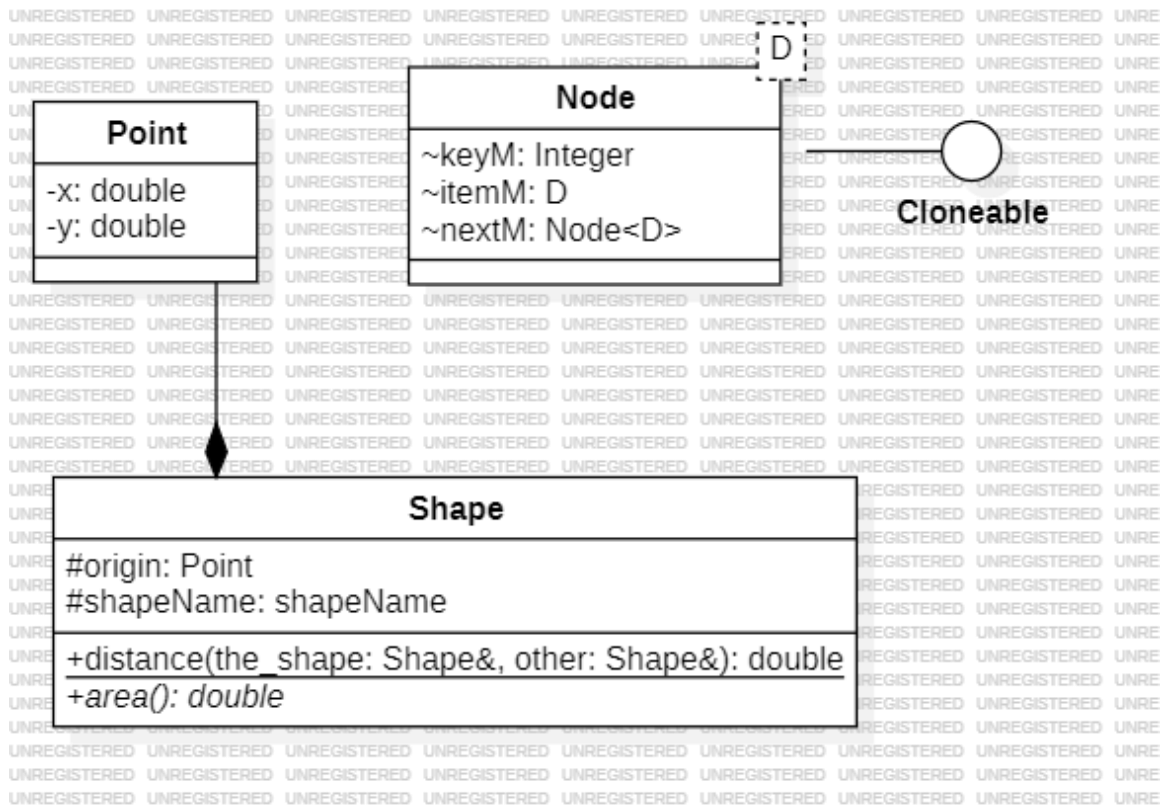
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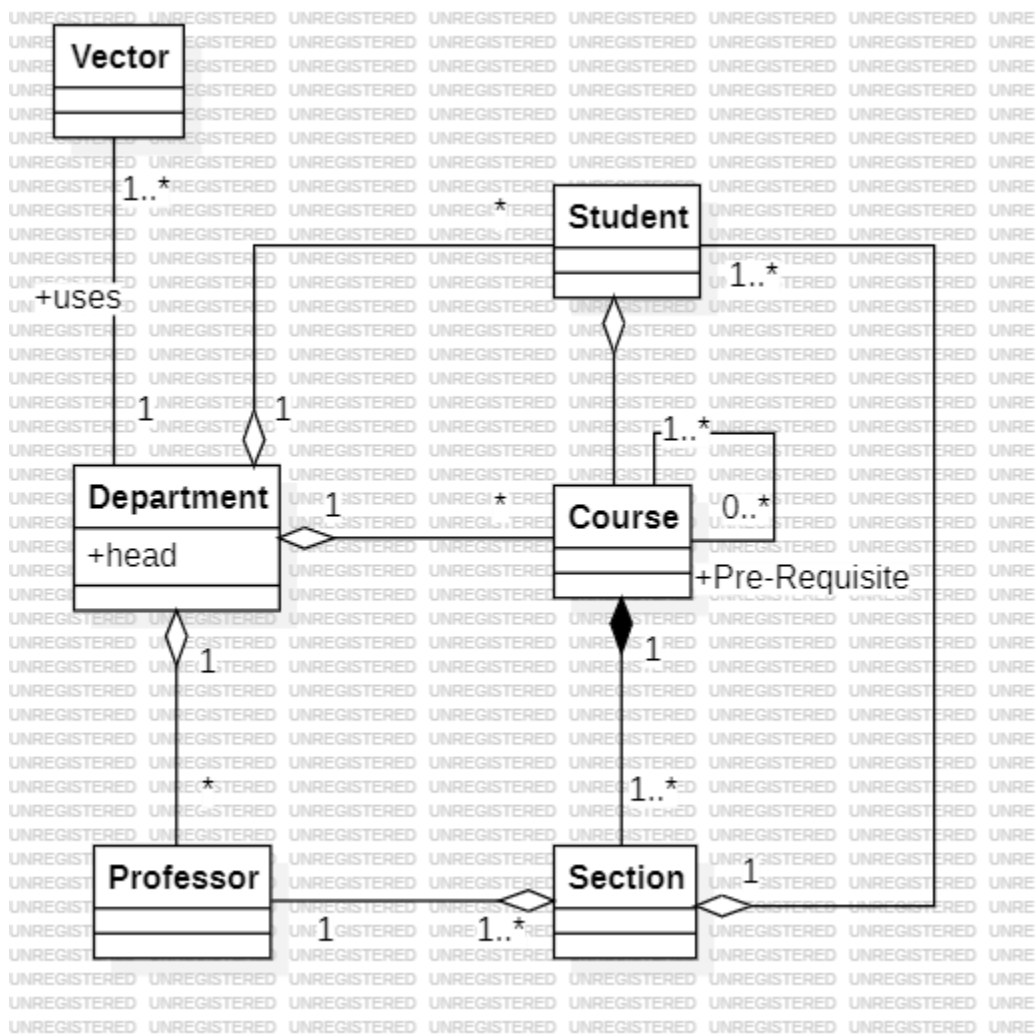
Lab Section: B02

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Exercise A



Exercise B



Exercise C

Moveable.h

```
#ifndef MOVEABLE_H
#define MOVEABLE_H

class Moveable
{
public:
    virtual void forward() = 0;
    virtual void backward() = 0;
};

#endif
```

Resizable.h

```
#ifndef RESIZEABLE_H
#define RESIZEABLE_H

class Resizable
{
public:
    virtual void enlarge(int n) = 0;
    virtual void shrink(int n) = 0;
};

#endif
```

Vehicle.h

```
#ifndef VEHICLE_H
#define VEHICLE_H

#include <string>

using namespace std;

#include "Moveable.h"
#include "Resizable.h"

class Vehicle : public Moveable, public Resizable
{
public:
    Vehicle(string name);
    virtual void move() = 0;

protected:
    string name;
};

#endif
```

Car.h

```
#ifndef CAR_H
#define CAR_H

#include <string>

#include "Vehicle.h"

class Car final: public Vehicle
{
public:
    void turn();

private:
    int seats;
};

#endif
```

Exercise D

(Please check exD.png for full resolution image)

